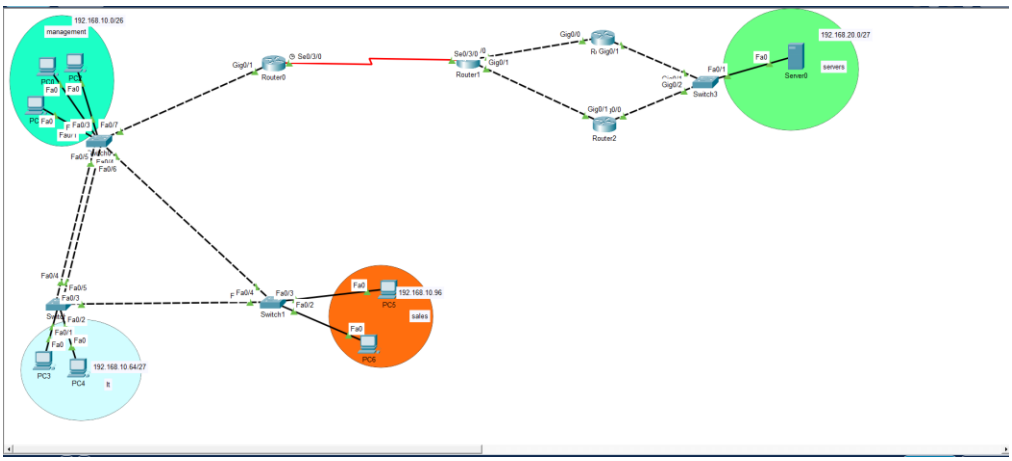
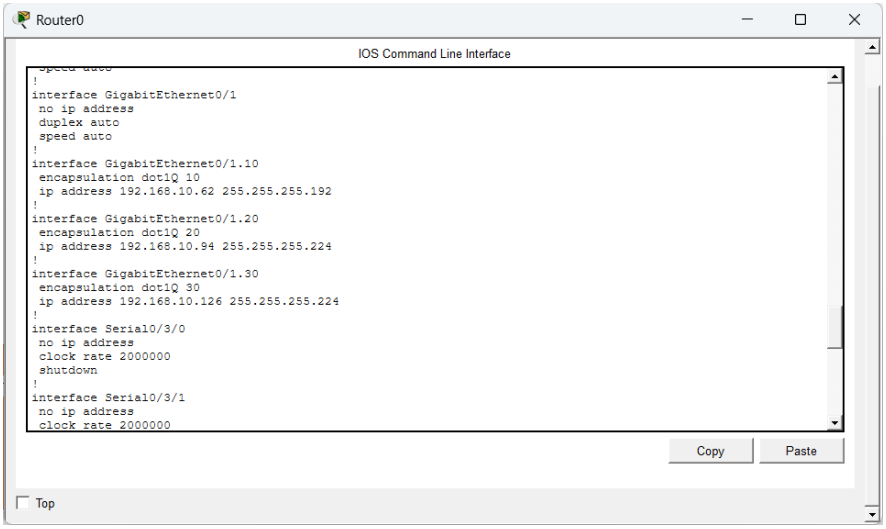
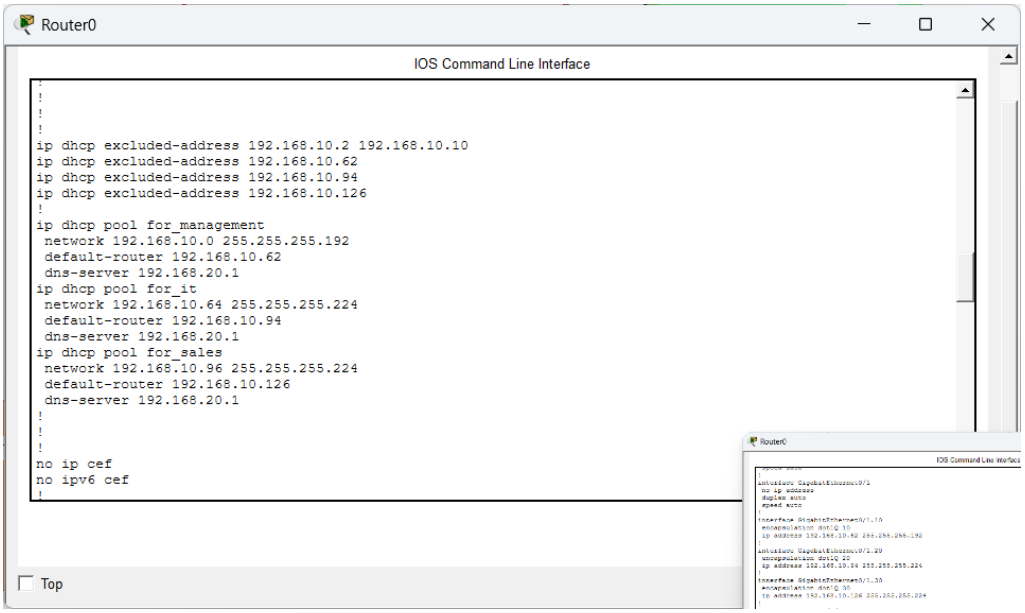
Enterprise Network Design with VLAN Segmentation, DHCP, FHRP, DNS, and ACL Implementation



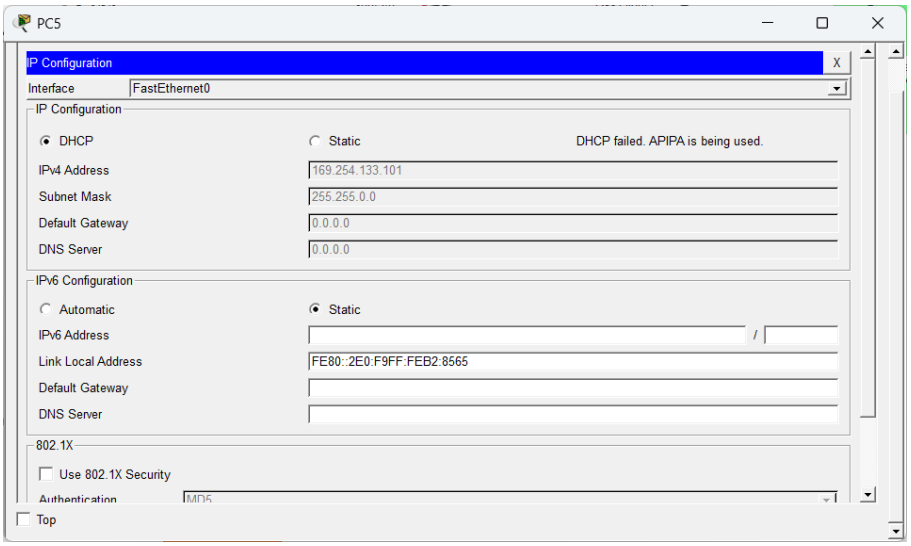
This topology represents a small enterprise network divided into Management, IT, and sales. VLANs were implemented to separate departments, while trunk links were configured between switches to carry VLAN traffic. Router-on-a-stick was used for inter-VLAN routing, and OSPF was configured for routing across the WAN links.

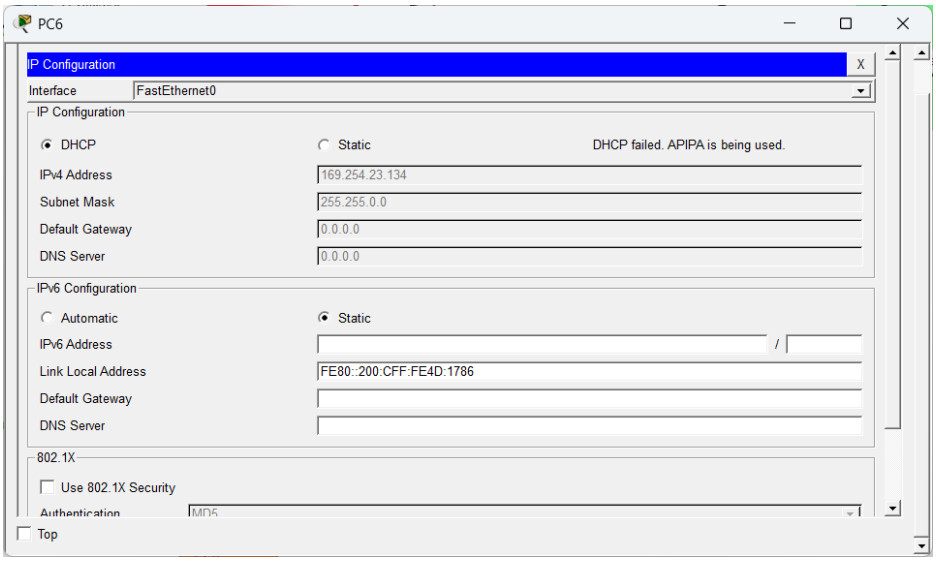


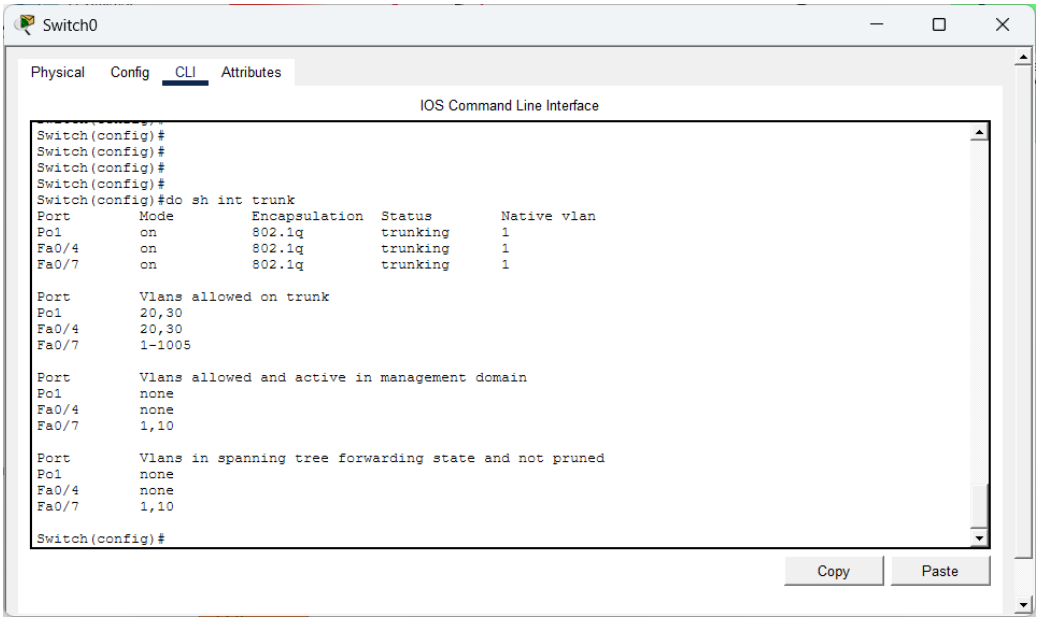
Router-on-a-stick was used for inter-VLAN routing with the default gateways for each VLAN configured using router sub-interfaces.

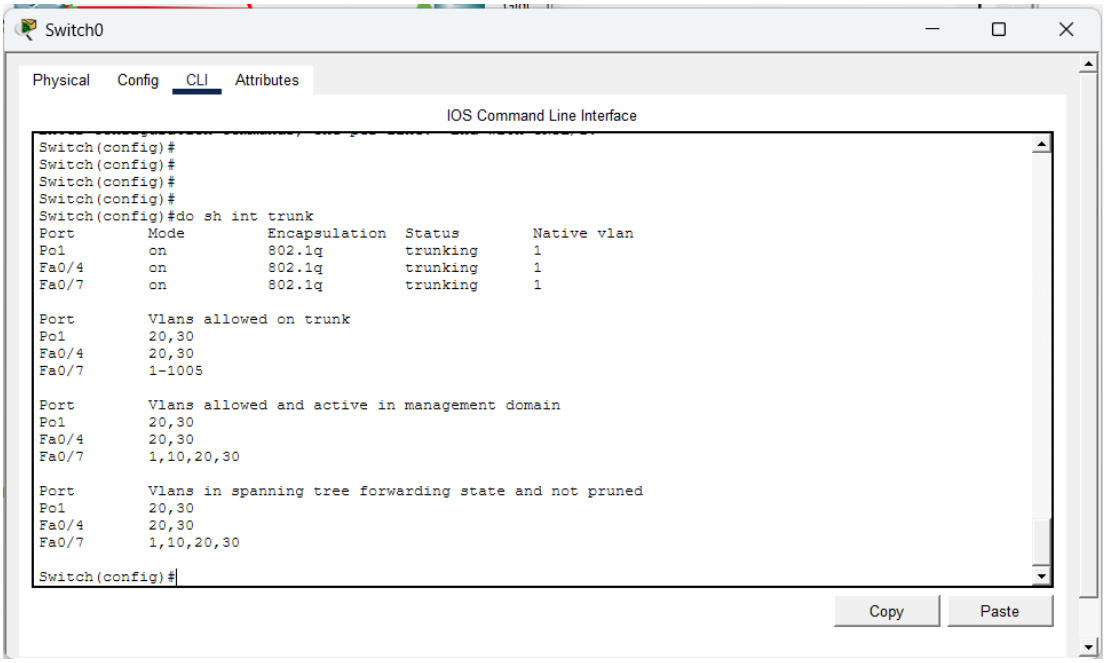


DHCP pools were created for each VLAN, and gateway IP addresses were excluded from DHCP pools to prevent address conflicts.




PC6
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
DHCP failed. APIPA is being used.
IPv4 Address
169.254.23.134
Subnet Mask
255.255.0.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::200:CFF:FE4D:1786
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Top


Switch0
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#do sh int trunk
Port Mode Encapsulation Status Native vlan
Po1 on 802.1q trunking 1
Fa0/4 on 802.1q trunking 1
Fa0/7 on 802.1q trunking 1
Port Vlans allowed on trunk
Po1 20,30
Fa0/4 20,30
Fa0/7 1-1005
Port Vlans allowed and active in management domain
Po1 none
Fa0/4 none
Fa0/7 1,10
Port Vlans in spanning tree forwarding state and not pruned
Po1 none
Fa0/4 none
Fa0/7 1,10
Switch(config)#
Copy
Paste


Switch0
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Config
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IOS Command Line Interface
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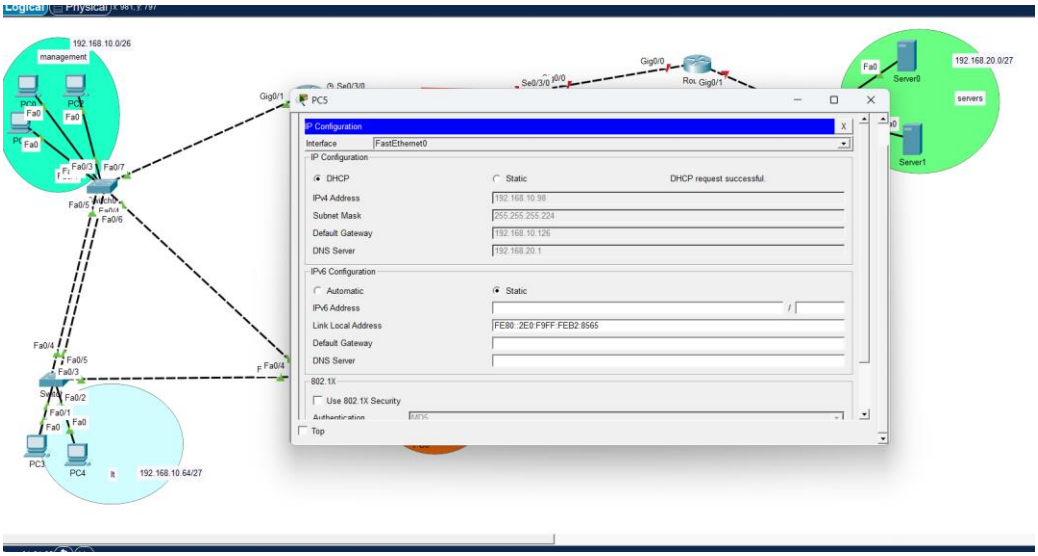
Fa0/4 on 802.1q trunking 1
Fa0/7 on 802.1q trunking 1
Port Vlans allowed on trunk
Po1 20,30
Fa0/4 20,30
Fa0/7 1-1005
Port Vlans allowed and active in management domain
Po1 20,30
Fa0/4 20,30
Fa0/7 1,10,20,30
Port Vlans in spanning tree forwarding state and not pruned
Po1 20,30
Fa0/4 20,30
Fa0/7 1,10,20,30
Switch(config)#
Copy
Paste


192.168.10.0/26
management
192.168.20.0/27
servers
Server0
Server1
PC5
IP Configuration
Interface FastEthernet0
DHCP
Static
DHCP request successful.
IPv4 Address 192.168.10.98
Subnet Mask 255.255.255.224
Default Gateway 192.168.10.126
DNS Server 192.168.20.1
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address FE80::2E0:F9FF:FEB2:8565
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Top
192.168.10.64/27

Router0

IOS Command Line Interface

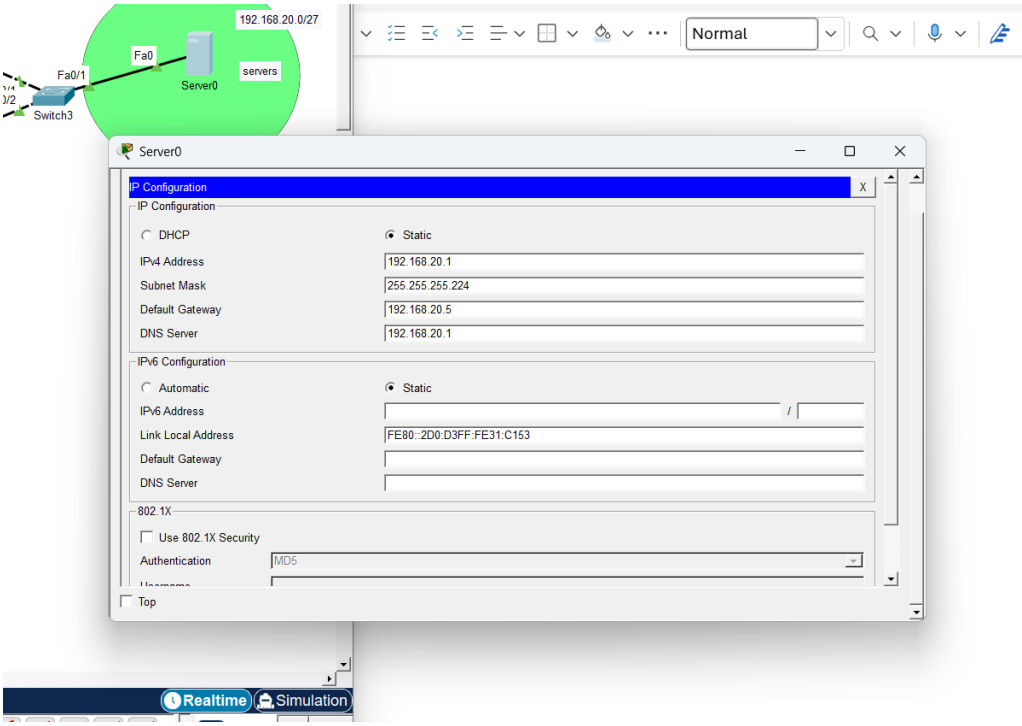
```
      * - candidate default, U - per-user static route, o - ODR
      P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.10.0.0/30 is directly connected, Serial0/3/0
L       10.10.0.1/32 is directly connected, Serial0/3/0
O       10.10.0.4/30 [110/65] via 10.10.0.2, 00:41:08, Serial0/3/0
O       10.10.0.8/30 [110/65] via 10.10.0.2, 00:42:36, Serial0/3/0
     192.168.10.0/24 is variably subnetted, 6 subnets, 3 masks
C       192.168.10.0/26 is directly connected, GigabitEthernet0/1.10
L       192.168.10.62/32 is directly connected, GigabitEthernet0/1.10
C       192.168.10.64/27 is directly connected, GigabitEthernet0/1.20
L       192.168.10.94/32 is directly connected, GigabitEthernet0/1.20
C       192.168.10.96/27 is directly connected, GigabitEthernet0/1.30
L       192.168.10.126/32 is directly connected, GigabitEthernet0/1.30
     192.168.20.0/27 is subnetted, 1 subnets
O       192.168.20.0/27 [110/66] via 10.10.0.2, 00:29:11, Serial0/3/0

Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
```

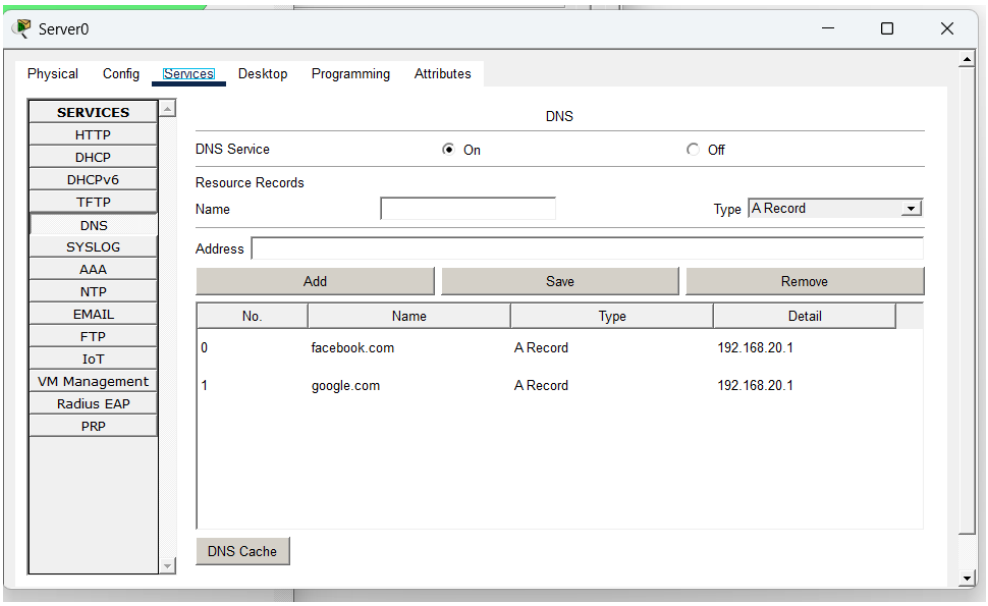
Copy Paste

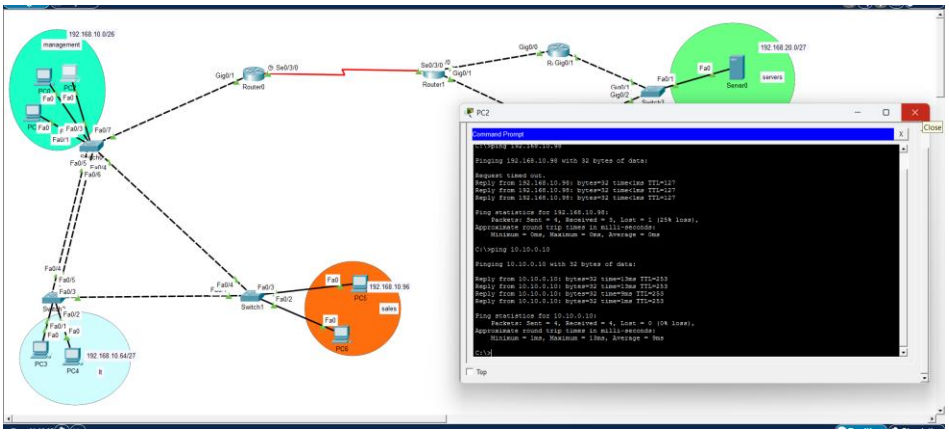
Top

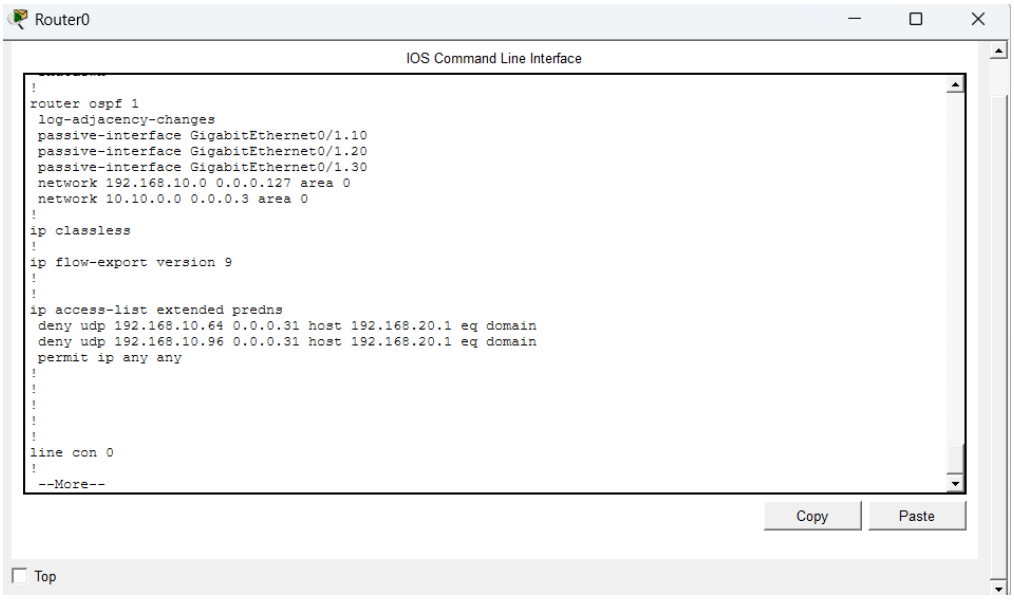
```
Router(config)#do sh standby
GigabitEthernet0/0 - Group 1 (version 2)
  State is Active
    10 state changes, last state change 02:19:25
  Virtual IP address is 192.168.20.5
  Active virtual MAC address is 0000.0C9F.F001
    Local virtual MAC address is 0000.0C9F.F001 (v2 default)
  Hello time 3 sec, hold time 10 sec
    Next hello sent in 1.264 secs
  Preemption enabled
  Active router is local
  Standby router is 192.168.20.3
  Priority 123 (configured 123)
  Group name is hsrp-Gig0/0-1 (default)
Router(config)#
Router(config)#
Router(config)#
```

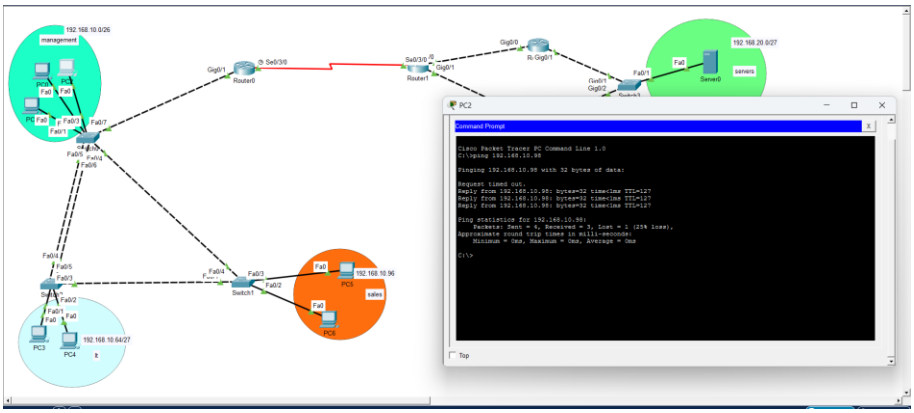
HSRP was implemented to provide redundancy for the default gateway in the server network.


192.168.20.0/27
Fa0
servers
Server0
Fa0/1
Switch3
Normal
Server0
IP Configuration
IP Configuration
DHCP
Static
IPv4 Address
192.168.20.1
Subnet Mask
255.255.255.224
Default Gateway
192.168.20.5
DNS Server
192.168.20.1
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::2D0:D3FF:FE31:C153
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Top
Realtime
Simulation


Server0
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
DNS
DNS Service
On
Off
Resource Records
Name
Type
A Record
Address
Add
Save
Remove
No.
Name
Type
Detail
0
facebook.com
A Record
192.168.20.1
1
google.com
A Record
192.168.20.1
DNS Cache

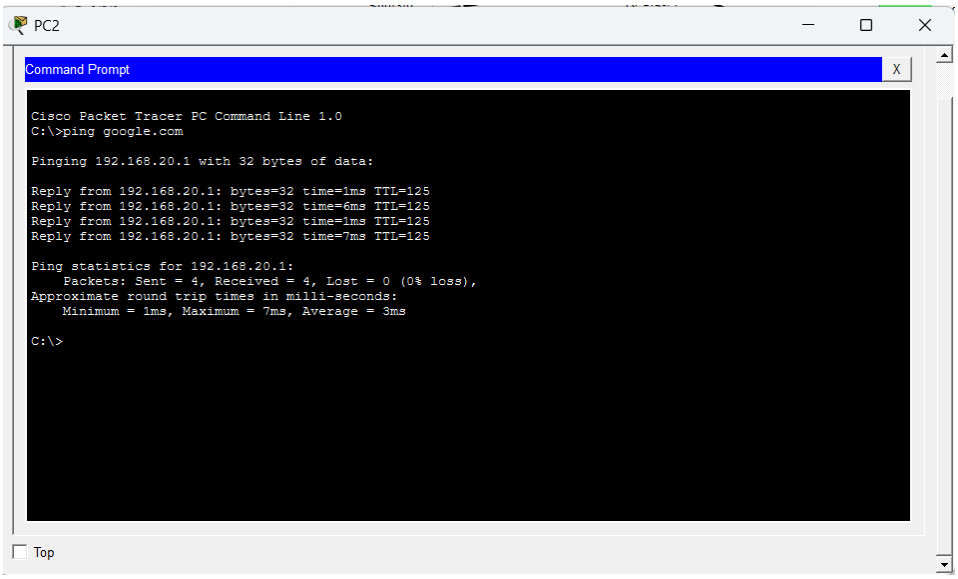

PC2
Command Prompt
Close
Top

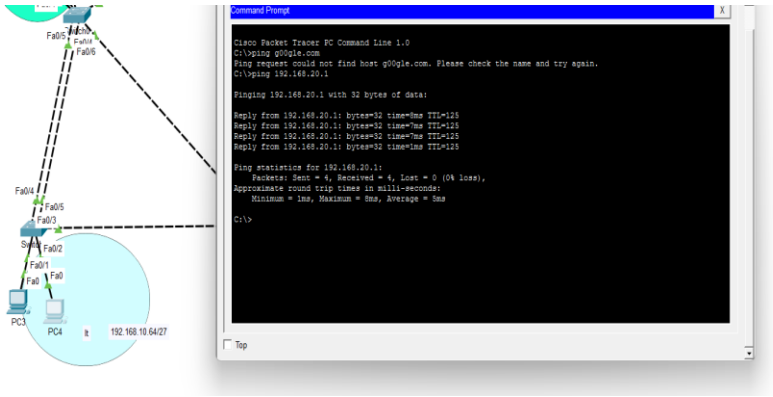




An extended ACL was configured on Router0 to restrict access to simulated internet services, allowing only the Management VLAN.

```
C:\>nslookup google.com

Server:  [192.168.20.1]
Address:  192.168.20.1

Non-authoritative answer:
Name:    google.com
Address:   192.168.20.1

C:\>
```





When the user from the IT department tries to access google.com, ICMP traffic was successfully blocked by the ACL, verifying correct policy enforcement.

Summary:

The initial 192.168.10.0/24 Ip was assigned to the company by ISP which was further subnetted into 3 subnets 192.168.10.0/26 for management (“VLAN10”) with 50 hosts,

192.168.10.64/27 for IT department (“VLAN20”) with 25 hosts, and 192.168.10.96/27 for Sales department (“VLAN30”) with 19 hosts. All the switches are connected to each other for redundancy purposes and are load balanced through VLAN IDs. For example, SWITCH2 is the root switch for VLAN30 but SWITCH0 is the secondary root switch for it, in same way all the VLANS are load balanced. An EtherChannel connection is done between switch 0 and switch 2 to improve the redundancy. Further link-state routing protocol “OSPF” is used as a routing protocol for the design, and FHRP (“First Hop Redundancy Protocol”) is used to provide redundancy for the default gateway of the server. A DNS server was deployed in a separate server network (`192.168.20.0/27`), which is reachable through OSPF routing across the WAN links. For WAN routing, 10.10.0.0/30 were used.

Finally, extended ACLs were implemented to enforce network security policies, restricting access to selected services so that only the Management VLAN could access simulated internet resources.

Problem Faced During Configuration:

During the DHCP configuration, it was observed that only devices in directly connected VLANs were receiving IP addresses, while hosts in remote VLANs failed to obtain IP assignments.

Upon investigation, the issue was identified as incorrect trunk configuration between switches, which prevented VLAN traffic — including DHCP broadcast messages — from being forwarded across the network. As a result, DHCP requests from remote VLANs were not reaching the router acting as the DHCP server.

After properly configuring the trunk links and ensuring that all required VLANs were allowed across inter-switch connections, the issue was resolved, and all devices successfully received IP addresses.

Key learning outcomes:

This project enhanced understanding of VLAN segmentation, DHCP broadcast behavior, inter-VLAN routing using sub-interfaces, and the importance of correct trunk configuration. It also demonstrated how multiple technologies such as OSPF, HSRP, DNS, and ACLs integrate to form a secure and scalable enterprise network.